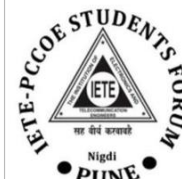


PCET'S  
Pimpri Chinchwad College Of  
Engineering, Nigdi - Pune  
Electronics & Telecommunication  
Student  
Association- ISF, IEEE SPS & IETE



# NEXUS 2026

## Aero-X

**Event Type :** Technical Event (Workshop + Competition)

**Venue :-** E&TC Department, PCCOE

**Date :-** 9<sup>th</sup> and 10<sup>th</sup> October 2026

**Event Coordinators :-**

| TY Coordinators  | SY Coordinators  |
|------------------|------------------|
| V Taraksh Prijay | Manthan Waghmare |
| Chinmayi Pethkar | Arya Jadhav      |
| Manan Gandhi     |                  |

## Description:-

### Event Overview:

Aero-X is a two-day hands-on technical workshop and competition on drone technology, conducted in association with Prarambh-X Technologies. Participants learn the fundamentals of flight, aerodynamics, drone components and DGCA regulations, then assemble, calibrate and fly their own quadcopter. Advanced concepts such as flight controller configuration, GPS and compass integration and mission planning are covered on professional-grade drone kits.

Learning is put to the test through a technical quiz and a Drone Arena obstacle course challenge. The event encourages teamwork, problem-solving and the practical application of engineering concepts.

## **Problem Statement: Build and Fly Your Own Quadcopter**

Drones are now used across agriculture, surveying, defence and delivery, yet most engineers have never built one from the ground up. Understanding how a drone flies requires more than theory: it means knowing how every component works together and what goes wrong when it does not.

In teams, assemble a quadcopter from individual components, calibrate it, and fly it through an obstacle course within the constraints given on site.

## **OBJECTIVES & LEARNING OUTCOMES**

### **Primary Objective**

To give participants practical, end-to-end exposure to drone technology, from the physics of flight to assembling, calibrating and piloting a working quadcopter.

### **Participants will gain experience in:**

- **Principles of flight:** lift, drag, thrust and weight, and how drones control roll, pitch and yaw
- **Drone anatomy:** motors, ESCs, flight controllers, batteries, propellers, transmitters, receivers and sensors
- **Hands-on assembly:** motor direction, propeller orientation, binding and gyro calibration of a quadcopter
- **Advanced systems:** Pixhawk flight controller setup, GPS and compass integration and mission planning
- **Piloting:** simulator-based pilot training and live flying
- **Regulations and safety:** DGCA drone rules, Ground Control Stations and LiPo battery safety
- **Teamwork:** problem-solving and decision-making under competition conditions

## **Components & Hardware Provided in Kit:-**

### **1. Give-Away Kit: DIY Quadcopter (one per team registered with kit)**

Each eligible team will receive a basic quadcopter kit containing the essential hardware required for assembly, configuration, testing, and operation

The kit will generally include:

- Quadcopter frame and structural components
- Flight-control system with stabilization and receiver functionality
- Motors and compatible propellers
- Electronic control and power-management components
- Rechargeable battery and charging system
- Remote-control system
- Required wiring, connectors, and assembly accessories

### **2. Advanced Training Kit: Quadcopter and Hexacopter (for training only)**

An advanced multirotor platform will be used during the workshop to demonstrate flight-control systems, navigation, propulsion, calibration, and advanced drone operations.

The training setup may include:

- Advanced multirotor frame and flight-control system
- Brushless motors, electronic speed controllers, and propellers
- Power-management and battery-monitoring system
- GPS, compass, and other flight sensors
- Safety, monitoring, and flight-logging components
- Required wiring, connectors, and supporting accessories

**\*Note:** The advanced training kit is provided exclusively for demonstration, training, testing, and supervised practical sessions and will not be distributed to participants.

## Event Schedule:-

| Time                 | Session                                                                                                                                                                                             | Duration |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>DAY 01</b>        |                                                                                                                                                                                                     |          |
| 10:00 to 10:05 AM    | <b>All Aboard:</b> Orientation of the workshop                                                                                                                                                      | 5 Min    |
| 10:05 to 10:20 AM    | <b>The Evolution of Flight and Drone Types:</b> From the first hot air balloon and the Wright brothers to modern autonomous UAVs                                                                    | 15 Min   |
| 10:20 to 11:20 AM    | <b>Principles of Aerodynamics:</b> Lift, drag, thrust and weight, Bernoulli's principle, Newton's laws, control surfaces and roll, pitch, yaw. Demos: Tug-of-Forces with glider, FPV manoeuvres     | 60 Min   |
| 11:20 to 11:30 AM    | <b>Pit Stop:</b> Short break                                                                                                                                                                        | 10 Min   |
| 11:30 AM to 12:45 PM | <b>Drone Anatomy, From Motors to Sensors:</b> Motors, ESCs, batteries, propellers, radios, flight controllers and sensors. Activities: Spin the Motor, One-Arm Demo, Obstacle Avoidance Challenge   | 75 Min   |
| 12:45 to 1:30 PM     | <b>Lunch Break</b>                                                                                                                                                                                  | 45 Min   |
| 1:30 to 2:30 PM      | <b>Hexa Masterclass (Advanced Kit):</b> Soldering, wiring, Pixhawk configuration, calibration and mission planning on S500 / hexacopter platforms                                                   | 60 Min   |
| 2:30 to 3:00 PM      | <b>Technical Quiz (50 Marks):</b> Aviation history, principles of flight, drone components and assembly. Top 10 teams qualify for the Drone Arena                                                   | 30 Min   |
| 3:00 to 5:00 PM      | <b>Drone Simulator Training:</b> Hands-on transmitter handling and pilot training on the simulator                                                                                                  | 120 Min  |
| 5:00 to 6:00 PM      | <b>DGCA Rules and Regulations:</b> Drone Rules, Ground Control Station, Mission Planner, battery safety. Followed by announcement of the top 10 teams                                               | 60 Min   |
| <b>DAY 02</b>        |                                                                                                                                                                                                     |          |
| 9:00 to 11:00 AM     | <b>Assembly and Piloting Session:</b> Teams assemble their DIY quadcopter from individual components, set correct motor and propeller orientation, bind and calibrate, and fly the drone they built | 120 Min  |
| 11:00 AM to 12:30 PM | <b>Drone Arena:</b> Obstacle course flying competition for the top 10 teams                                                                                                                         | 90 Min   |
| 12:30 to 1:00 PM     | <b>Airshow, Winners Announcement and Closing Ceremony:</b> Demonstration flights of high-performance drones and gliders, followed by prize distribution                                             | 30 Min   |

\*Schedule is tentative and may change.

## Prize Pool:-

**Total Prize Pool:** ₹25,000+

- **1st Prize :** ₹12,000
- **2nd Prize:** ₹8,000
- **3rd Prize :** ₹5,000

\*Terms and conditions apply.

## Event Rules and Regulations:-

### 1. Team & Participation

- Participation is allowed in teams of 2 to 5 members.
- **Registration options:**  
Teams may register:
  - With Kit (one DIY quadcopter kit per team, which the team keeps).
  - Without Kit (full workshop and competition participation, no take-home kit).
- Participants registering individually will be asked to form a team with other participants at the venue, or will be assigned to a team, as the event involves group activities.
- Team composition must remain the same across both days and all rounds.

### 2. Workshop Participation

- Participants are expected to attend all sessions on both days.
- Advanced training kits are for demonstration only.
- Instructions of Prarambh-X trainers must be followed during assembly, calibration and flying.

### 3. Competition

- **Technical Quiz:** a 50-mark quiz for all teams. The top 10 teams qualify for the Drone Arena.
- **Drone Arena:** qualified teams fly a drone through an obstacle course. The team that crosses all obstacles and completes the course within the constraints announced on site will be declared the winner.
- Teams registered with kit will fly the DIY quadcopter they assembled. Teams registered without kit will fly a drone provided by the organisers.

- Only minor adjustments such as tightening, trimming and propeller replacement are allowed. No additional components or external modifications are permitted.
- Drones may be flown only inside the designated arena and only when instructed by the arena marshal.
- Judges' decisions will be final and binding.

#### **4. Safety**

- All kits operate on **low-voltage DC LiPo batteries (3.7V to 12.6V)**.
- No high-voltage equipment is involved.
- Keep hands and face clear of propellers whenever a drone is powered. Bench work on advanced kits is done with propellers removed.
- LiPo batteries must be charged, handled and stored only as instructed by trainers.
- Participants are responsible for reasonable care of their kits during the sessions.
- Standard PCCOE lab safety protocols apply.
- Prarambh X staff trained in first aid will be present.

#### **Evaluation Criteria:-**

##### **Technical Quiz**

- Total marks: 50
- Shortlisting based on marks scored, with time taken as the tie-breaker
- Top 10 teams qualify for the Drone Arena
- The quiz has 3 papers-15 marks , 15 marks , and 10 marks.
- The quiz is to be solved by the team together. It's not an individual quiz.

##### **Drone Arena**

- Teams fly through an obstacle course within the constraints (time limit and flight rules) announced on site
- The team that crosses all obstacles and completes the course within the constraints is the winner
- Crashes, touching obstacles or flying outside the arena boundary may attract penalties as announced on site
- Tie-breakers, if needed, will be announced on site

## **FAQs:-**

### **1. Is prior drone-building experience required?**

No, beginners are welcome.

### **2. Will the kit be provided?**

Teams registered With Kit receive one DIY quadcopter kit per team, which they keep. Teams registered Without Kit take part in all sessions but do not receive a take-home kit. Additional kits can be purchased at the end of the workshop.

### **3. Can I register individually?**

Yes. You will be asked to team up with other participants at the venue, or will be assigned to a team.

### **4. Can we modify the drone before the competition?**

Only minor adjustments are allowed.